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Inventor(s)

Zhuang; Mingzhe et al.

Providing information for locating an item within a warehouse from a shopper to other shoppers retrieving the item from the warehouse

Abstract

Based on orders fulfilled by shoppers of an online concierge system, the online concierge system identifies items in an order that are difficult to find in a warehouse in which the order is fulfilled. When a shopper obtains a difficult to find item from the warehouse, the online concierge system prompts the shopper to provide information for finding the difficult to find item in the warehouse. The online concierge system stores the information for finding the difficult to find item from the shopper in association with the difficult to find item and with the warehouse. Subsequently, when a different shopper is fulfilling an order from the warehouse including the difficult to find item, the online concierge system displays the information for finding the difficult to find item in the warehouse to the different shopper.

Inventors: Zhuang; Mingzhe (Los Altos Hills, CA), van Horne; Camille (San Francisco, CA), Rudnick; Christopher (Pacifica, CA), Knight; Benjamin (Oakland, CA), Jenkins; Chris (Thornton, CO), Andonova; Viktoriya (San Francisco, CA), Gluhovic; Djordje (Oakland, CA), Sejpal; Riddhima (Dublin, CA), Golivkin; Maksim (Oakland, CA), Rao Karikurve; Sharath (Berkeley, CA)

Applicant: Maplebear Inc. (San Francisco, CA)

Family ID: 71404457

Assignee: Maplebear Inc. (San Francisco, CA)

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Primary Examiner: Georgalas; Anne M

Attorney, Agent or Firm: Fenwick & West LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of co-pending U.S. application Ser. No. 16/734,273, filed Jan. 3, 2020, which claims the benefit of U.S. Provisional Application No. 62/788,726, filed Jan. 4, 2019, which is incorporated by reference in its entirety.

BACKGROUND

(1) This disclosure relates generally to a process for a shopper retrieving an item from a warehouse, and more specifically to providing information locating certain items within the warehouse from a shopper to other shoppers.

(2) In current online concierge systems, shoppers (or "pickers") fulfill orders at a physical warehouse, such as a retailer, on behalf of customers as part of an online shopping concierge service. In current online concierge systems, the shoppers may be sent to various warehouses with instructions to fulfill orders for items, and the pickers then find the items included in the customer order in a warehouse. But in conventional online concierge systems it is difficult to know before a shopper arrives at the warehouse if the item in the customer's order is in stock at the warehouse. Item inventory may fluctuate throughout a day or week, such that even if a shopper previously found an item at a warehouse, the shopper may be unable to find the item at the same warehouse for a subsequent delivery order.

(3) Further, certain items may be available at a warehouse but difficult for a shopper to locate within the warehouse. For example, an item may be located in a location within a warehouse that is difficult for a shopper to see, or a shopper's view of the item within the warehouse may be obscured by other items or other physical obstructions within the warehouse. When a shopper fulfills an

order including one or more items that are difficult to find within the warehouse, the shopper may spend unnecessary time looking for a difficult to locate item within the warehouse, increasing an amount of time for the shopper to fulfill the order and provide the items in the order to the customer. Additionally, a shopper's inability to find a difficult to locate item within a warehouse may cause the shopper to request the customer select an alternative item in place of the difficult to locate item, increasing customer frustration by having the user select an alternative item when the originally requested item is available from the warehouse, but difficult to locate within the warehouse.

SUMMARY

(4) An online concierge system can generate and use a machine-learned model to predict item availability of items included in a delivery order and selected by a customer. The machine-learned model is trained using information about items, such as whether the items were found at a warehouse in previous orders. The previous delivery orders make up large scale training datasets that are used to statistically map item characteristics, order information, and other factors to item availability within the machine-learned model. Item information from new orders is then input into the machine-learned model to generate item availability probabilities. Based on the availability predictions from the machine-learned model, instructions are generated to a shopper who fulfills a delivery order. The instructions may reduce the amount of time that a shopper spends looking for an item at a warehouse by telling the shopper that an item is likely to be available or unavailable, and instructing the shopper to continue or stop looking for an item based on the predicted availability.

(5) While the machine-learned model predicts availability of an item at a warehouse, an item that is available at the warehouse but difficult to be located within the warehouse by a shopper. For example, an item may be obscured from view within a warehouse by a physical obstruction or by other items within the warehouse. As another example, an item may be located in a section of the warehouse with unrelated objects or may be located in an area of the warehouse that is distant from other similar items. This may prevent various shoppers from finding and retrieving an item in the warehouse, even though the item is actually available at the warehouse.

(6) To allow shoppers to better identify certain items within a warehouse that are difficult to locate within the warehouse, when the online concierge system receives an order from a customer including one or more items to be fulfilled at a warehouse, the online concierge system identifies a difficult to find item within the order. In various embodiments, an item is “difficult to find” if the trained model predicts that the item has at least a threshold availability at the warehouse and that a shopper other than the shopper fulfilling the order was unable to retrieve from the warehouse within a threshold amount of time from when the online concierge system received the order. For example, a difficult to find item is an item an order to be fulfilled at a warehouse that at least one shopper other than the shopper fulfilling the order did not find within the warehouse within 24 hours of a time from a time when the online concierge system received the order.

(7) In response to receiving an indication that the shopper fulfilling the order including the difficult to find item at the warehouse retrieving the difficult to find item from the warehouse, the online concierge system prompts the shopper to provide information for finding the difficult to find item within the warehouse. For example, the online concierge system receives an indication from a shopper mobile application executing on a client device of the shopper identifying the difficult to find product, the warehouse, the shopper, and an indication that the shopper has retrieved the difficult to find product from the warehouse location. In the preceding example, the online concierge system transmits a prompt to the shopper mobile application that prompts the shopper to specify information for finding the difficult to find item within the warehouse. The prompt may include a message congratulating the shopper for locating the difficult to find item within the warehouse. In various embodiments, the information for finding the difficult to find item is a picture of a location of the difficult to find item within the warehouse, a text description of a

location of the difficult to find item within the warehouse, directions for locating the difficult to find item within the warehouse, or any combination thereof. The online concierge system stores the information for finding the difficult to find item within the warehouse in association with an identifier of the warehouse and in association with an identifier of the difficult to find item. In some embodiments, the online concierge system also stores an identifier of the shopper who retrieved the difficult to find item in association with the information for finding the difficult to find item within the warehouse.

(8) Subsequently, when another shopper is fulfilling an order at the warehouse that includes the item that had been designated “difficult to find,” the online concierge system retrieves the information for finding the difficult to find item within the warehouse stored in association with the identifier of the warehouse and the identifier of the difficult to find item. Via the shopper mobile application, the online concierge system displays the information for finding the difficult to find item within the warehouse to the other shopper fulfilling the order at the warehouse. For example, the online concierge system displays an indication that a location hint is available proximate to information identifying the difficult to find item in an interface. In response to receiving a selection of the indication from the shopper mobile application, the online concierge system displays the information for finding the difficult to find item within the warehouse, allowing the other shopper to leverage the stored information to more efficiently locate and retrieve the difficult to find item from the warehouse.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 illustrates an environment of an online shopping concierge service, according to one embodiment.

(2) FIG. 2 is a diagram of an online shopping concierge system, according to one embodiment.

(3) FIG. 3A is a diagram of a customer mobile application (CMA), according to one embodiment.

(4) FIG. 3B is a diagram of a shopper mobile application (SMA), according to one embodiment.

(5) FIG. 4 is a flowchart of a process for predicting inventory availability, according to one embodiment.

(6) FIG. 5 is a flowchart of a process for updating training datasets for a machine-learned model, according to one embodiment.

(7) FIG. 6 is a flowchart of a process for determining instructions to a shopper if a probability indicates that an item is available at a warehouse, according to one embodiment.

(8) FIG. 7 is a flowchart of a process for determining feedback to a customer based on a probability that an item is available at a warehouse, according to one embodiment.

(9) FIG. 8 is a flowchart of a process for obtaining information for finding certain items within a warehouse for display to one or more shoppers, according to one embodiment.

(10) FIG. 9 shows an example interface displayed by the shopper mobile application displaying a prompt to a shopper to provide information for finding a difficult to find item within the warehouse, according to one embodiment.

(11) FIG. 10 shows an example interface displaying an order being fulfilled by a shopper, according to one embodiment.

(12) FIG. 11 shows an example interface displayed by the shopper mobile application including information for finding a difficult to find item in a warehouse, according to one embodiment.

(13) The figures depict embodiments of the present disclosure for purposes of illustration only. One skilled in the art will readily recognize from the following description that alternative embodiments of the structures and methods illustrated herein may be employed without departing from the principles, or benefits touted, of the disclosure described herein.

DETAILED DESCRIPTION

System Overview

(14) FIG. 1 illustrates an environment **100** of an online platform, according to one embodiment. The figures use like reference numerals to identify like elements. A letter after a reference numeral, such as “110a,” indicates that the text refers specifically to the element having that particular reference numeral. A reference numeral in the text without a following letter, such as “110,” refers to any or all of the elements in the figures bearing that reference numeral. For example, “110” in the text refers to reference numerals “110a” and/or “110b” in the figures.

(15) The environment **100** includes an online concierge system **102**. The system **102** is configured to receive orders from one or more customers **104** (only one is shown for the sake of simplicity). An order specifies a list of goods (items or products) to be delivered to the customer **104**. The order also specifies the location to which the goods are to be delivered, and a time window during which the goods should be delivered. In some embodiments, the order specifies one or more retailers from which the selected items should be purchased. The customer may use a customer mobile application (CMA) **106** to place the order; the CMA **106** is configured to communicate with the online concierge system **102**.

(16) The online concierge system **102** is configured to transmit orders received from customers **104** to one or more shoppers **108**. A shopper **108** may be a contractor, employee, or other person (or entity) who is enabled to fulfill orders received by the online concierge system **102**. The shopper **108** travels between a warehouse and a delivery location (e.g., the customer's home or office). A shopper **108** may travel by car, truck, bicycle, scooter, foot, or other mode of transportation. In some embodiments, the delivery may be partially or fully automated, e.g., using a self-driving car. The environment **100** also includes three warehouses **110a**, **110b**, and **110c** (only three are shown for the sake of simplicity; the environment could include hundreds of warehouses). The warehouses **110** may be physical retailers, such as grocery stores, discount stores, department stores, etc., or non-public warehouses storing items that can be collected and delivered to customers. Each shopper **108** fulfills an order received from the online concierge system **102** at one or more warehouses **110**, delivers the order to the customer **104**, or performs both fulfillment and delivery. In one embodiment, shoppers **108** make use of a shopper mobile application **112** which is configured to interact with the online concierge system **102**.

(17) FIG. 2 is a diagram of an online concierge system **102**, according to one embodiment. The online concierge system **102** includes an inventory management engine **202**, which interacts with inventory systems associated with each warehouse **110**. In one embodiment, the inventory management engine **202** requests and receives inventory information maintained by the warehouse **110**. The inventory of each warehouse **110** is unique and may change over time. The inventory management engine **202** monitors changes in inventory for each participating warehouse **110**. The inventory management engine **202** is also configured to store inventory records in an inventory database **204**. The inventory database **204** may store information in separate records—one for each participating warehouse **110**—or may consolidate or combine inventory information into a unified record. Inventory information includes both qualitative and quantitative information about items, including size, color, weight, SKU, serial number, and so on. In one embodiment, the inventory database **204** also stores purchasing rules associated with each item, if they exist. For example, age-restricted items such as alcohol and tobacco are flagged accordingly in the inventory database **204**. Additional inventory information useful for predicting the availability of items may also be stored in the inventory database **204**. For example, for each item-warehouse combination (a particular item at a particular warehouse), the inventory database **204** may store a time that the item was last found, a time that the item was last not found (a shopper looked for the item but could not find it), the rate at which the item is found, and the popularity of the item.

(18) Inventory information provided by the inventory management engine **202** may supplement the training datasets **220**. Inventory information provided by the inventory management engine **202**

may not necessarily include information about the outcome of picking a delivery order associated with the item, whereas the data within the training datasets **220** is structured to include an outcome of picking a delivery order (e.g., if the item in an order was picked or not picked).

(19) The online concierge system **102** also includes an order fulfillment engine **206** which is configured to synthesize and display an ordering interface to each customer **104** (for example, via the customer mobile application **106**). The order fulfillment engine **206** is also configured to access the inventory database **204** in order to determine which products are available at which warehouse **110**. The order fulfillment engine **206** may supplement the product availability information from the inventory database **204** with an item availability predicted by the machine-learned item availability model **216**. The order fulfillment engine **206** determines a sale price for each item ordered by a customer **104**. Prices set by the order fulfillment engine **206** may or may not be identical to in-store prices determined by retailers (which is the price that customers **104** and shoppers **108** would pay at the retail warehouses). The order fulfillment engine **206** also facilitates transactions associated with each order. In one embodiment, the order fulfillment engine **206** charges a payment instrument associated with a customer **104** when he/she places an order. The order fulfillment engine **206** may transmit payment information to an external payment gateway or payment processor. The order fulfillment engine **206** stores payment and transactional information associated with each order in a transaction records database **208**.

(20) In some embodiments, the order fulfillment engine **206** also shares order details with warehouses **110**. For example, after successful fulfillment of an order, the order fulfillment engine **206** may transmit a summary of the order to the appropriate warehouses **110**. The summary may indicate the items purchased, the total value of the items, and in some cases, an identity of the shopper **108** and customer **104** associated with the transaction. In one embodiment, the order fulfillment engine **206** pushes transaction and/or order details asynchronously to retailer systems. This may be accomplished via use of webhooks, which enable programmatic or system-driven transmission of information between web applications. In another embodiment, retailer systems may be configured to periodically poll the order fulfillment engine **206**, which provides detail of all orders which have been processed since the last request.

(21) The order fulfillment engine **206** may interact with a shopper management engine **210**, which manages communication with and utilization of shoppers **108**. In one embodiment, the shopper management engine **210** receives a new order from the order fulfillment engine **206**. The shopper management engine **210** identifies the appropriate warehouse to fulfill the order based on one or more parameters, such as a probability of item availability determined by a machine-learned item availability model **216**, the contents of the order, the inventory of the warehouses, and the proximity to the delivery location. The shopper management engine **210** then identifies one or more appropriate shoppers **108** to fulfill the order based on one or more parameters, such as the shoppers' proximity to the appropriate warehouse **110** (and/or to the customer **104**), his/her familiarity level with that particular warehouse **110**, and so on. Additionally, the shopper management engine **210** accesses a shopper database **212** which stores information describing each shopper **108**, such as his/her name, gender, rating, previous shopping history, and so on. Methods that can be used to identify a warehouse **110** at which a shopper **108** can likely find most or all items in an order are described with respect to FIGS. 4-7.

(22) As part of fulfilling an order, the order fulfillment engine **206** and/or shopper management engine **210** may access a customer database **214** which stores information describing each customer. This information could include each customer's name, address, gender, shopping preferences, favorite items, stored payment instruments, and so on.

Machine Learning Model

(23) The online concierge system **102** further includes a machine-learned item availability model **216**, a modeling engine **218**, and training datasets **220**. The modeling engine **218** uses the training datasets **220** to generate the machine-learned item availability model **216**. The machine-learned

item availability model **216** can learn from the training datasets **220**, rather than follow only explicitly programmed instructions. The inventory management engine **202**, order fulfillment engine **206**, and/or shopper management engine **210** can use the machine-learned item availability model **216** to determine a probability that an item is available at a warehouse **110**. The machine-learned item availability model **216** may be used to predict item availability for items being displayed to or selected by a customer, or included in received delivery orders. A single machine-learned item availability model **216** is used to predict the availability of any number of items.

(24) The machine-learned item availability model **216** can be configured to receive as inputs information about an item, the warehouse for picking the item, and the time for picking the item. The machine-learned item availability model **216** may be adapted to receive any information that the modeling engine **218** identifies as indicators of item availability. At minimum, the machine-learned item availability model **216** receives information about an item-warehouse pair, such as an item in a delivery order and a warehouse at which the order could be fulfilled. Items stored in the inventory database **204** may be identified by item identifiers. As described above, various characteristics, some of which are specific to the warehouse (e.g., a time that the item was last found in the warehouse, a time that the item was last not found in the warehouse, the rate at which the item is found, the popularity of the item) may be stored for each item in the inventory database **204**. Similarly, each warehouse may be identified by a warehouse identifier and stored in a warehouse database along with information about the warehouse. A particular item at a particular warehouse may be identified using an item identifier and a warehouse identifier. In other embodiments, the item identifier refers to a particular item at a particular warehouse, so that the same item at two different warehouses is associated with two different identifiers. For convenience, both of these options to identify an item at a warehouse are referred to herein as an “item-warehouse pair.” Based on the identifier(s), the online concierge system **102** can extract information about the item and/or warehouse from the inventory database **204** and/or warehouse database, and provide this extracted information as inputs to the item availability model **216**.

(25) The machine-learned item availability model **216** contains a set of functions generated by the modeling engine **218** from the training datasets **220** that relate the item, warehouse, and timing information, and/or any other relevant inputs, to the probability that the item is available at a warehouse. Thus, for a given item-warehouse pair, the machine-learned item availability model **216** outputs a probability that the item is available at the warehouse. The machine-learned item availability model **216** constructs the relationship between the input item-warehouse pair, timing, and/or any other inputs and the availability probability (also referred to as “availability”) that is generic enough to apply to any number of different item-warehouse pairs. In some embodiments, the probability output by the machine-learned item availability model **216** includes a confidence score. The confidence score may be the error or uncertainty score of the output availability probability, and may be calculated using any standard statistical error measurement. In some examples, the confidence score is based in part on whether the item-warehouse pair availability prediction was accurate for previous delivery orders (e.g., if the item was predicted to be available at the warehouse and not found by the shopper, or predicted to be unavailable but found by the shopper). In some examples, the confidence score is based in part on the age of the data for the item, e.g., if availability information has been received within the past hour, or the past day. The set of functions of the item availability model **216** may be updated and adapted following retraining with new training datasets **220**. The machine-learned item availability model **216** may be any machine learning model, such as a neural network, boosted tree, gradient boosted tree or random forest model. In some examples, the machine-learned item availability model **216** is generated from XGBoost algorithm.

(26) The item probability generated by the machine-learned item availability model **216** may be used to determine instructions delivered to the customer **104** and/or shopper **108**, as described in further detail below.

(27) The training datasets **220** relate a variety of different factors to known item availabilities from the outcomes of previous delivery orders (e.g. if an item was previously found or previously unavailable). The training datasets **220** include the items included in previous delivery orders, whether the items in the previous delivery orders were picked, warehouses associated with the previous delivery orders, and a variety of characteristics associated with each of the items (which may be obtained from the inventory database **204**). Each piece of data in the training datasets **220** includes the outcome of a previous delivery order (e.g., if the item was picked or not). The item characteristics may be determined by the machine-learned item availability model **216** to be statistically significant factors predictive of the item's availability. For different items, the item characteristics that are predictors of availability may be different. For example, an item type factor might be the best predictor of availability for dairy items, whereas a time of day may be the best predictive factor of availability for vegetables. For each item, the machine-learned item availability model **216** may weight these factors differently, where the weights are a result of a “learning” or training process on the training datasets **220**. The training datasets **220** are very large datasets taken across a wide cross section of warehouses, shoppers, items, warehouses, delivery orders, times and item characteristics. The training datasets **220** are large enough to provide a mapping from an item in an order to a probability that the item is available at a warehouse. In addition to previous delivery orders, the training datasets **220** may be supplemented by inventory information provided by the inventory management engine **202**. In some examples, the training datasets **220** are historic delivery order information used to train the machine-learned item availability model **216**, whereas the inventory information stored in the inventory database **204** include factors input into the machine-learned item availability model **216** to determine an item availability for an item in a newly received delivery order. In some examples, the modeling engine **218** may evaluate the training datasets **220** to compare a single item's availability across multiple warehouses to determine if an item is chronically unavailable. This may indicate that an item is no longer manufactured. The modeling engine **218** may query a warehouse **110** through the inventory management engine **202** for updated item information on these identified items.

Machine Learning Factors

(28) The training datasets **220** include a time associated with previous delivery orders. In some embodiments, the training datasets **220** include a time of day at which each previous delivery order was placed. Time of day may impact item availability, since during high-volume shopping times, items may become unavailable that are otherwise regularly stocked by warehouses. In addition, availability may be affected by restocking schedules, e.g., if a warehouse mainly restocks at night, item availability at the warehouse will tend to decrease over the course of the day. Additionally, or alternatively, the training datasets **220** include a day of the week previous delivery orders were placed. The day of the week may impact item availability, since popular shopping days may have reduced inventory of items, or restocking shipments may be received on particular days. In some embodiments, training datasets **220** include a time interval since an item was previously picked in a previously delivery order. If an item has recently been picked at a warehouse, this may increase the probability that it is still available. If there has been a long time interval since an item has been picked, this may indicate that the probability that it is available for subsequent orders is low or uncertain. In some embodiments, training datasets **220** include a time interval since an item was not found in a previous delivery order. If there has been a short time interval since an item was not found, this may indicate that there is a low probability that the item is available in subsequent delivery orders. And conversely, if there is has been a long time interval since an item was not found, this may indicate that the item may have been restocked, and is available for subsequent delivery orders. In some examples, training datasets **220** may also include a rate at which an item is typically found by a shopper at a warehouse, a number of days since inventory information about the item was last received from the inventory management engine **202**, a number of times an item was not found in a previous week, or any number of additional rate or time information. The

relationships between this time information and item availability are determined by the modeling engine **218** training a machine learning model with the training datasets **220**, producing the machine-learned item availability model **216**.

(29) The training datasets **220** include item characteristics. In some examples, the item characteristics include a department associated with the item. For example, if the item is yogurt, it is associated with the dairy department. The department may be the bakery, beverage, nonfood and pharmacy, produce and floral, deli, prepared foods, meat, seafood, dairy, the meat department, or dairy department, or any other categorization of items used by the warehouse. The department associated with an item may affect item availability, since different departments have different item turnover rates and inventory levels. In some examples, the item characteristics include an aisle of the warehouse associated with the item. The aisle of the warehouse may affect item availability, since different aisles of a warehouse may be more frequently re-stocked than others. Additionally, or alternatively, the item characteristics include an item popularity score. The item popularity score for an item may be proportional to the number of delivery orders received that include the item. An alternative or additional item popularity score may be provided by a retailer through the inventory management engine **202**. In some examples, the item characteristics include a product type associated with the item. For example, if the item is a particular brand of a product, then the product type will be a generic description of the product type, such as “milk” or “eggs.” The product type may affect the item availability, since certain product types may have a higher turnover and re-stocking rate than others, or may have larger inventories in the warehouses. In some examples, the item characteristics may include a number of times a shopper was instructed to keep looking for the item after he or she was initially unable to find the item, a total number of delivery orders received for the item, whether or not the product is organic, vegan, gluten free, or any other characteristics associated with an item. The relationships between item characteristics and item availability are determined by the modeling engine **218** training a machine learning model with the training datasets **220**, producing the machine-learned item availability model **216**.

(30) The training datasets **220** may include additional item characteristics that affect the item availability, and can therefore be used to build the machine-learned item availability model **216** relating the delivery order for an item to its predicted availability. The training datasets **220** may be periodically updated with recent previous delivery orders. The training datasets **220** may be updated with item availability information provided directly from shoppers **108**, as described in further detail with reference to FIG. 5. Following updating of the training datasets **220**, a modeling engine **218** may retrain a model with the updated training datasets **220**, and produce a new machine-learned item availability model **216**.

Customer Mobile Application

(31) FIG. 3A is a diagram of the customer mobile application (CMA) **106**, according to one embodiment. The CMA **106** includes an ordering interface **302**, which provides an interactive interface with which the customer **104** can browse through and select products and place an order. The CMA **106** also includes a system communication interface **304** which, among other functions, receives inventory information from the online shopping concierge system **102** and transmits order information to the system **102**. The CMA **106** also includes a preferences management interface **306** which allows the customer **104** to manage basic information associated with his/her account, such as his/her home address and payment instruments. The preferences management interface **306** may also allow the user to manage other details such as his/her favorite or preferred warehouses **110**, preferred delivery times, special instructions for delivery, and so on.

Shopper Mobile Application

(32) FIG. 3B is a diagram of the shopper mobile application (SMA) **112**, according to one embodiment. The SMA **112** includes a barcode scanning module **320** which allows a shopper **108** to scan an item at a warehouse **110** (such as a can of soup on the shelf at a grocery store). The barcode scanning module **320** may also include an interface which allows the shopper **108** to

manually enter information describing an item (such as its serial number, SKU, quantity and/or weight) if a barcode is not available to be scanned. SMA **112** also includes a basket manager **322** which maintains a running record of items collected by the shopper **108** for purchase at a warehouse **110**. This running record of items is commonly known as a “basket”. In one embodiment, the barcode scanning module **320** transmits information describing each item (such as its cost, quantity, weight, etc.) to the basket manager **322**, which updates its basket accordingly. The SMA **112** also includes a system communication interface **324** which interacts with the online shopping concierge system **102**. For example, the system communication interface **324** receives an order from the system **102** and transmits the contents of a basket of items to the system **102**. The SMA **112** also includes an image encoder **326** which encodes the contents of a basket into an image. For example, the image encoder **326** may encode a basket of goods (with an identification of each item) into a QR code which can then be scanned by an employee of the warehouse **110** at check-out.

Predicting Inventory Availability

(33) As described with reference to FIG. 2, the machine-learned item availability model **216** of the online concierge system **102** can determine an availability of an item requested by the customer **104**. FIG. 4 is a flowchart illustrating a process **400** for predicting inventory availability, according to one embodiment. The online concierge system **102** receives **402** a delivery order that includes a set of items and a delivery location. The delivery location may be any location associated with a customer, such as a customer's home or office. The delivery location may be stored with the customer location in the customer database **214**. Based on the delivery order, the online concierge system **102** identifies a warehouse **404** for picking the set of items in the delivery order based on the set of items and the delivery location. In some cases, the customer specifies a particular warehouse or set of warehouses (e.g., a particular grocery store or chain of grocery stores) in the order. In other cases, the online concierge system **102** selects the warehouse based on the items and the delivery location. In some examples, there are a number of different possible warehouses that the set of items may be picked from. The warehouses may be identified by the order fulfillment engine **206** based on warehouses stored by the inventory management engine **202**, and warehouses are identified with a suitable inventory and within a threshold distance of the delivery address. In some embodiments, a single delivery order can be split into multiple orders and picked at multiple warehouses, e.g., if the items cannot be fulfilled at a single warehouse. In this example, each possible warehouse is input into the machine-learned item availability model **216**.

(34) After the warehouses are identified, the online concierge system **102** retrieves **406** the machine-learned item availability model **216** that predicts a probability that an item is available at the warehouse. The items in the delivery order and the identified warehouses are input into the machine-learned item availability model **216**. For example, the online concierge system **102** may input the item, warehouse, and timing characteristics for each item-warehouse pair into the machine-learned item availability model **216** to assess the availability of each item in the delivery order at each potential warehouse at a particular day and/or time. The machine-learned item availability model **216** predicts **408** the probability that one of the set of items in the delivery order is available at the warehouse. If a number of different warehouses are identified **404**, then the machine-learned item availability model **216** predicts the item availability for each one. In some examples, the probability that an item is available includes a probability confidence score generated by the machine-learned item availability model **216**.

(35) The order fulfillment engine **206** uses the probability to generate **410** an instruction to a shopper. The order fulfillment engine **206** transmits the instruction to the shopper through the SMA **112** via the shopper management engine **210**. The instruction is based on the predicted probability. In some examples, the shopper management engine **210** instructs the shopper to pick an item in the delivery order at a warehouse with the highest item availability score. For example, if a warehouse is more likely to have more items in the delivery order available than another warehouse, then the

shopper management engine **210** instructs the shopper to pick the item at the warehouse with better availability. Other examples of the shopper management engine **210** instruction to the shopper are described in further detail with reference to FIGS. **5** and **6**. In some other examples, the order fulfillment engine **206** sends a message and/or instruction to a customer based on the probability predicted by the machine-learned item availability model **216**. This is described in further detail with reference to FIG. **7**.

Updating the Training Datasets

(36) FIG. **5** is a flowchart illustrating a process **500** for updating training datasets for a machine-learned model, according to one embodiment. The training datasets may be the training datasets **220** as shown in FIG. **2**. While the training datasets **220** include large datasets of information collected from previous delivery orders (e.g., information identifying items and whether the items were available at a warehouse), certain items or warehouses might have less information associated with them in the training datasets **220** than other items or warehouses. For example, if an item is not frequently ordered, or has not been ordered for a long period of time, then it may be more difficult to build an accurate availability prediction in the machine-learned item availability model **216**. One way to improve the ability of the machine-learned item availability model **216** to accurately predict item availability is to increase the information about the item in the training datasets **220**, and add new information. With larger and/or more recent datasets on the item, the modeling engine **218** can build more statistically meaningful connections between the machine-learning factors described with reference to FIG. **2** and the predicted item availability.

(37) Process **500** thus improves the machine-learned item availability model **216** by increasing the datasets for particular items in the training datasets **220** with low confidence scores. Process **500** may be carried out by the online concierge system **102**, e.g., by the inventory management engine **202** in conjunction with the shopper management engine **210**, the item availability model **216**, and the modeling engine **218**. In some examples, process **500** is carried out by the online concierge system **102** following retrieving **406** a machine-learned model that predicts a probability that an item is available at a warehouse, as described in FIG. **4**.

(38) The online concierge system **102** (e.g., the inventory management engine **202** using the item availability model **216**) identifies **502** an item-warehouse pair. For example, the item and warehouse in the item-warehouse pair may be an item in a received order and warehouse or potential warehouse for picking the items from the received order, e.g., to evaluate the suitability of the warehouse or likelihood of successfully picking the order before the order is picked.

(39) As another example, the item-warehouse pair may be identified from items for which the availability predicted by the machine-learned item availability model **216** was incorrect (e.g., the item was predicted to be available and was determined by the shopper to be out of stock, or the item was predicted to be unavailable and the shopper was able to find it in the warehouse). For items for which the availability prediction was incorrect, the online concierge system **102** may determine if the items have sufficient associated information within the training datasets **220**. If the online concierge system **102** determines that the incorrect probability was a result of insufficient or stale information in the training datasets **220**, it may identify item-warehouse pairs and carry out process **500** to update the training datasets **220**.

(40) Additionally, or alternatively, item-warehouse pairs are identified from new items offered by the online concierge system **102**. For new items, there may not be previous delivery order information relating the item availability to item characteristics, delivery order information, or time information in the training datasets **220**. The lack of previous delivery orders may lead to a low confidence score for new items. The inventory management engine **202** may initiate the process **500** for new items until sufficient information about the items are collected in the training datasets **220** to improve the item availability confidence score associated with the items.

(41) The online concierge system **102** (e.g., the inventory management engine **202** using the machine-learned item availability model **216**) inputs the item, warehouse, and timing

characteristics for the identified item-warehouse pair into the machine-learned item availability model **216** and determines **504** a confidence score associated with a probability that an item is available at the warehouse. The online concierge system **102** may determine probabilities and/or confidence scores for all or selected items in an inventory, e.g., items that are expected to be picked based on already-received orders, sales, promotions, holidays, weather, historical trends, or other factors. The confidence score is generated along with the item availability probability (also referred to as “availability”) by the machine-learned item availability model **216**. The confidence score may be an error associated with the availability probability. The confidence score indicates items that may not have enough training data in the training datasets **220** to generate a statistically significant link between the item's availability and information from the delivery order and/or item characteristics. In some alternate embodiments, the online concierge system **102** may identify, using the item availability model **216**, item-warehouse pairs with a low confidence score, e.g., all item-warehouse pairs with a confidence score below a particular threshold. This list of item-warehouse pairs may be filtered, e.g., based on item popularity, predicted items to be ordered, warehouse, or one or more other factors.

(42) In response to the determined confidence level of an item-warehouse pair being below a threshold, the online concierge system **102** (e.g., the shopper management engine **210**) instructs **506** the shopper to collect new information about items with a confidence score below a threshold. A confidence score threshold may be an item availability probability between 0 and 1. A threshold confidence score may be 0.3, such that in response to a confidence score below 0.3, the shopper is instructed to collect new information about an item. In some embodiments, the online concierge system **102** also considers the availability probability for the item-warehouse pair. For example, if an item-warehouse pair has a confidence level slightly below the threshold, but a very low or very high availability probability, the online concierge system **102** may determine not to collect new information about the item-warehouse pair. In some embodiments, the threshold used for the confidence score may depend on the availability probability, or vice versa.

(43) In response to the instruction, the shopper **108** determines whether the item is available at the warehouse. The shopper may be instructed to try to find the item at the warehouse, and indicate, through the SMA **112**, whether the item is available. This information is transmitted to the online concierge system **102** via the shopper management engine **210**, and used to update **508** the training datasets **220**. In some embodiments, a shopper may be given a list of items with low confidence scores to seek within the warehouse. The online concierge system **102** updates **508** the training dataset **220** with new information about the item, which includes whether or not the item is available in the warehouse, and any additional item characteristics, warehouse information, or time information as described with respect to FIG. 2. The online concierge system **102** also updates the inventory database **204** based on the received information; e.g., if the inventory database **204** stores the time at which the item was most recently found or not found, this time will be updated based on the input from the shopper **108**. In response to the new information collected by the shopper, the modeling engine **218** may update or retrain the machine learning item availability model **216** with the updated training datasets **220**. Process **500** may be carried out by the online concierge system **102** until a confidence score associated with a probability that an item is available is above a threshold.

Use Case Examples

(44) An example of process **500** used in conjunction with process **400** is described below. The online concierge system **102** receives **402** a delivery order from a customer **104** through the CMA **106**. The customer **104** schedules a delivery at their home of three items to be delivered the following day. As an example, the customer **104** may order grated mozzarella, pizza dough, and tomato sauce, each of which is included in the delivery order. The online concierge system **102** sends the delivery order to the order fulfillment engine **206**. The order fulfillment engine **206** uses the inventory management engine **202** and customer database **214** to identify **404** a warehouse for

picking the requested items based on the items and the delivery location (i.e., the customer's home). A number of possible warehouses may be identified. For each possible warehouse, the order fulfillment engine **206** identifies **502** an item-warehouse pair with one of the items in the delivery order. Thus, a set of item-warehouse pairs is identified for each of the grated mozzarella, pizza dough and tomato sauce. The online concierge system **102** retrieves **406** the machine-learned item availability model **216** that predicts a probability that an item is available at the warehouse. The online concierge system **102** inputs the item, warehouse, and timing characteristics for each of the identified item-warehouse pairs into the machine-learned item availability model **216**. The machine-learned item availability model **216** predicts **408** the probability that each of the grated mozzarella, pizza dough and tomato sauce are available at the identified warehouses. For each of the availability probabilities, the online concierge system **102** also determines **504** a confidence score associated with the probability from the machine-learned item availability model **216**.

(45) It is possible that the confidence score for pizza dough confidence score at one or more of the warehouses is below a threshold, given that people frequently make their own pizza dough and it may not be frequently ordered. Thus, pizza dough may have a relatively small and/or old associated dataset in the training dataset **220**, leading to a low confidence score on the pizza dough availability probability within the machine-learned item availability model **216**. The online concierge system **102**, using the shopper management engine **210**, instructs **506** a shopper to collect new information about pizza dough at one or more of the warehouses. The shopper management engine **210** may identify an off-duty shopper, or a shopper already at one of the warehouses identified **502** in an item-warehouse pair to collect information about whether or not pizza dough is available at the warehouse. The shopper management engine **210** transmits this instruction through the SMA **112**. The shopper **108** may find that pizza dough is in fact available, and transmit the availability to the online concierge system **102** through the SMA **112**. This new information is used to update **508** the training dataset **220** and the inventory database **204**. The shopper management engine **210** may transmit the same instruction to multiple shoppers **108** at different warehouses, or at different times, such that there is a larger set of data about pizza dough availability added to the training dataset **220**, and more recent data in the inventory database **204**.

(46) In this example, the modeling engine **218** uses the updated training datasets **220** to retrain the machine-learned item availability model **216**. The online concierge system **102** then re-inputs the pizza dough-warehouse pairs into the updated machine-learned item availability model **216** and determines **504** a confidence score associated with the probability that pizza dough is available at a number of possible warehouses. It is possible that the confidence scores are now above a threshold, because the increased data about pizza dough added to the training datasets **220** has improved the machine-learned item availability model **216**, and/or the newer data in the inventory database **204** has improved the confidence score. The online concierge system **102** then generates **410** an instruction to a shopper **108** based on the availability probabilities for pizza dough. The instruction may be to pick the pizza dough at the warehouse with the highest availability probability. In other examples, the instruction may be to pick the pizza dough, grated mozzarella and tomato sauce at a warehouse with the highest availability probability for all of these items in the customer's delivery order. The online concierge system **102** transmits the instruction to a mobile device of the shopper **108**.

(47) Additionally, or alternatively, the online concierge system **102** may use the machine-learned item availability model **216** to predict an anticipated demand for an item at a warehouse. The online concierge system **102** may compare the number of times an item is included in a set of delivery orders to the item availability predictions generated by the machine-learned item availability model **216**, and identify items that are frequently ordered but have low corresponding availability probabilities. For example, around the holidays, there may be an increase in delivery orders including Brussels sprouts, whereas Brussels sprouts may have a low availability prediction since they are not typically stocked in large quantities. The online concierge system may identify

the discrepancy between a large volume of item orders and the low availability probability and convey this information to a warehouse **110**. Additionally, or alternatively, the online concierge system **102** may transmit information about items that have availability predictions below a threshold.

Instructions to Shopper

(48) FIG. **6** is a flowchart illustrating a process **600** for determining instructions to a shopper if a probability indicates that an item is available at a warehouse, according to one embodiment. Process **600** may be used to assist a shopper looking for an item in a delivery order at a warehouse, and may therefore reduce the time a shopper spends looking for items that are not actually available at a warehouse. Process **600** may be carried out by the online concierge system **102**.

(49) The online concierge system **102** (e.g., the shopper management engine **210**) receives an indication **602** from a shopper that he or she cannot find an item at the warehouse. The shopper may transmit this information to the online concierge system **102** through the SMA **112**, which communicates it to the shopper management engine **210**. The shopper may input the item information into the SMA **112**. In some examples, the shopper may also provide additional information about where they have already looked for the item within the warehouse, such as aisles in which the item was not found, departments in which the item was not found, the amount of time he or she spent looking for the item, etc. In response, the online concierge system **102** inputs the item, warehouse, and timing characteristics for the item received from the shopper and the warehouse in which the shopper is unable to find the item into the machine-learned item availability model **216**. In some embodiments, the online concierge system **102** may incorporate the information provided by the shopper through the SMA **112** into the training datasets **220**, which may be later used by the modeling engine **218** to update the machine-learned item availability model **216**. The online concierge system **102** determines **604** a probability that the item is available at the warehouse from the probability output by the machine-learned item availability model **216**. The online concierge system **102** then compares the output probability against a threshold and determines **606** if the item availability probability is above the threshold. In some examples, this threshold value may be an item availability probability of 0.3, or 30%. Additionally, or alternatively, the online concierge system **102** may compare a confidence score associated with the item availability probability to a threshold value.

(50) If an availability probability is above the threshold, this indicates that the item is predicted to be available at the warehouse. The shopper management engine **210** then instructs **608** a shopper to continue looking for the item. The instruction may be transmitted to the shopper through the SMA **112**. In some examples, the instruction may be accompanied by information as to a location within the warehouse that the item is most likely to be available, such as an aisle of the warehouse and/or a department.

(51) If the probability that the item is available is below a threshold value, then the shopper management engine **210** instructs **610** the shopper to stop looking for the item. The shopper management engine **210** may transmit the instruction through the SMA **112**. The shopper management engine **210** may add the item-warehouse pair and any associated time or item information to the training dataset **220** indicating that the item was not found at the warehouse. The shopper management engine **210** may then instruct the shopper to look for the next item in a delivery order, or for a replacement item that has a high availability probability.

(52) In some examples, the online concierge system **102** may determine **604** a probability that an item is available at a warehouse and compare **606** the availability probability to a threshold before receiving an indication **602** from a shopper that he or she cannot find an item. For example, the inventory management engine **202** may determine item availability probabilities for all items within a delivery order transmitted to a shopper. If the probability indicates that an item should be available, the online concierge system **102** may provide this information to the shopper through the SMA **112**. If the probability indicates that an item might be unavailable, the online concierge

system **102** may transmit a warning or other indication to the shopper that the item might be unavailable. In some examples, if the item probability indicates that an item is unavailable, the SMA **112** may instruct the shopper to limit the amount of time the shopper looks for the item in the warehouse, and/or to pick a replacement item. In some examples, the item availability probabilities provided to the shopper may include location information, such as where in a warehouse the item is most likely to be located, such as an aisle or department.

(53) In some embodiments, when transmitting a delivery order to a shopper, the online concierge system **102** visually distinguishes items in the delivery order having less than a threshold availability probability from items having greater than the delivery order. For example, the online concierge system **102** transmits a delivery order to a client device (e.g., a mobile device) of a shopper along with instructions to display a set of items in the delivery order having less than the threshold availability probability identified in the instructions in a banner displayed above a list of items in the delivery order. In some embodiments, the online concierge system **102** ranks items having availability probabilities less than the threshold availability probability so items with lower availability probabilities have higher positions in the ranking. The online concierge system **102** transmits instructions to the client device of the shopper that display the items with less than the threshold availability probability in the banner in an order based on the ranking; hence, items with lower availability probabilities are initially displayed in the banner. In some embodiments, a single item is displayed in the banner, and when the shopper selects the banner, a shopper mobile application **112** executing on the client device navigates to a position of the item displayed in the banner in a list. In some embodiments, the item with less than the threshold availability probability is visually distinguished from other items in the list (e.g., displayed in a different color, displayed with a highlight, displayed with a prominent border from other items, etc.). The shopper mobile application **112** also updates the banner to display a different item having less than the threshold availability probability, allowing the banner to display a single item having less than the threshold availability probability at once and to display different items having less than the threshold availability probability in response to the shopper selecting the banner.

Feedback to Customer

(54) FIG. 7 is a flowchart illustrating a process **700** for determining feedback to a customer based on a probability that an item is available at a warehouse, according to one embodiment. Process **700** may be carried out by the online concierge system **102** (e.g., the order fulfillment engine **206**) communicating with a customer via the CMA **106**. The order fulfillment engine **206** provides a customer interface **702**. The customer interface includes an ordering interface through which a customer may make item selections, and add items to a delivery order. The customer interface may be an interface of the CMA **106**, such as the ordering interface **302** as described in FIG. 3A. The customer interface receives **704** an item to be included in a delivery order. This item may be any item selected for purchase by the customer through the customer interface. The customer may also provide a delivery time associated with the order, which the online concierge system **102** can use to determine or approximate a picking time for the order. In response to the customer inputs, the online concierge system **102** (e.g., the order fulfillment engine **206**) determines **706** a probability that the item received at **704** is available at a warehouse, e.g., a warehouse selected by the customer, or a warehouse selected by the online concierge system **102** for fulfilling an order from the customer. The probability is determined by inputting the item, warehouse, and timing characteristics for the item received and the warehouse into a machine-learned item availability model **216**. The machine-learned item availability model **216** then outputs a probability that the item is available at the warehouse.

(55) The order fulfillment engine **206** then determines **708** if the probability an item is available at the warehouse is below a threshold. In some examples, this threshold is a probability between 0.1 and 0.3. In some examples, the probability may also include a confidence score as provided by the machine-learned item availability model **216**, and order fulfillment engine **206** may also determine

if the confidence score associated with the probability **706** is above a threshold. If the probability that an item is available is not below a threshold, then the order fulfillment engine **206** allows **710** a customer to add the item to the delivery order. This delivery order may then be transmitted to a shopper through the SMA **112** to be picked at a warehouse.

(56) If the probability that an item is available is below a threshold, then the order fulfillment engine **206** notifies the customer **712** through an ordering interface of the customer interface provided. The notification may be a warning or other message transmitted to the customer through the ordering interface. For example, the notification may be a message saying “item frequently not found” provided through the ordering interface. The ordering interface provides alternative options **714** to the item to the user. The alternative options may be items determined by the machine-learned item availability model **216** to be available. The alternative options **714** may be items of the same item type selected by the user that have high availability probabilities. For example, if process **700** receives an order for a specific brand of eggs **704**, and the probability that the eggs are available at the warehouse is below a threshold **708**, then the alternative options **714** may be other brands of the same kind of egg previously selected by the user (e.g., organic, brown, extra large, etc.) with high availability probabilities as determined by the machine-learned item availability model **216**. The alternative options **714** may be ranked according to their availability probabilities. To encourage customers to select from the alternative options, a message may be included with the alternative options indicated that the item is likely available or was recently found at the warehouse.

(57) In some examples, while the customer is notified **712** and provided with alternative options **714**, the ordering interface may still allow the customer to add the item to the delivery order. The customer may instruct the shopper to substitute the item received with the alternative options provided if the shopper cannot find the item. In other examples, the ordering interface does not allow the customer to add the item received **704**, and the customer chooses one of the alternative options **714** (or no item) to be added to the delivery order.

(58) In some examples, if the online concierge system **102** frequently receives customer requests to add an item to a delivery order that is then determined to have an availability probability below a threshold, the online concierge system **102** may temporarily remove the item from the item options provided to a customer through the customer interface.

Displaying Information for Finding One or More Items in a Warehouse From a Shopper

(59) FIG. **8** is a flowchart of one embodiment of a process **800** for obtaining information for finding certain items within a warehouse for display to one or more shoppers. The process **800** may be carried out by the online concierge system **102** (e.g., the order fulfillment engine **206**) communicating with a customer via the SMA **108**. In various embodiments, the process **800** performs the steps described in conjunction with FIG. **8** in a different order than the order shown in FIG. **8**. Further, in some embodiments, the process **800** includes different or additional steps than those described in conjunction with FIG. **8**.

(60) While the online concierge system **102** predicts availability of an item at a warehouse, as further described above in conjunction with FIGS. **4-7**, an item may be available at the warehouse (i.e., have at least a threshold availability determined by the online concierge system **102**) but be difficult to locate within the warehouse by a shopper. For example, an item may be obscured from view within a warehouse by a physical obstruction or by other items within the warehouse. As another example, an item may be located in a section of the warehouse with unrelated objects or may be located in an area of the warehouse that is distant from other similar items. This may prevent various shoppers from finding and retrieving an item in the warehouse, even though the item is available at the warehouse.

(61) To allow shoppers to better identify certain items within a warehouse that are difficult to locate within the warehouse, the online concierge system **102** identifies **805** a difficult to find item (also referred to as a “specific item”) within a warehouse from stored order information. In some

embodiments, the online concierge system **102** identifies **805** a difficult to find item as an item having at least a threshold availability at the warehouse and that a shopper was unable to retrieve from the warehouse within a specific time interval. In some embodiments, the online concierge system **102** identifies **805** a difficult to find item when the online concierge system **102** receives an order from a user to be fulfilled at the warehouse. For example, the online concierge system **102** identifies **805** a difficult to find item within the warehouse as an item in the received order having at least a threshold availability at the warehouse and that a shopper different than the shopper fulfilling the received order was unable to find in the warehouse within a specific time interval (e.g., 24 hours) of a time when the online concierge system **102** received the order. As another example, the online concierge system **102** retrieves information describing previously completed orders from the warehouse and identifies **805** a difficult to find item as an item with at least a threshold availability at the warehouse that was not found by at least a threshold number of shoppers at the warehouse within a specific time interval (e.g., 24 hours) of a time when the online concierge system **102** retrieved the information describing the previously completed orders. In the preceding examples, the online concierge system **102** determines the availability of the difficult to find item at the warehouse by inputting the difficult to find item, the warehouse, and timing characteristics from the order into a machine-learned item availability model, as further described above in conjunction with FIGS. 2-7. The machine-learned item availability model predicts the probability that the difficult to find item is available at the warehouse, as further described above in conjunction with FIGS. 2-6. The online concierge system **102** stores information identifying the difficult to find item in association with information identifying the warehouse; this allows the online concierge system **102** to maintain a record of one or more difficult to find items corresponding to the warehouse. As different warehouses may have different difficult to find items, maintaining an association between a difficult to find item and a warehouse allows the online concierge system **102** to identify difficult to find items in different warehouses.

(62) As shoppers fulfill orders, the online concierge system **102** receives **810** information from shopper mobile applications **112** corresponding to items the shoppers obtain from different warehouses to fulfill orders. The information the online concierge system **102** receives **810** from a shopper mobile application **112** identifies the shopper, identifies a warehouse, and identifies one or more items the shopper obtained from the warehouse. For example, the online concierge system **102** receives a shopper identifier, a warehouse identifier, and one or more item identifiers from the shopper mobile application **112** that the shopper obtained from the warehouse corresponding to a shopper.

(63) The online concierge system **102** compares information identifying items a shopper obtained from the warehouse to one or more identifiers of difficult to find items associated with the warehouse. In response to determining **815** a shopper obtained a difficult to find item associated with the warehouse from the warehouse, the online concierge system **102** prompts **820** the shopper who obtained the difficult to find item from the warehouse to provide information for finding the difficult to find item within the warehouse. In some embodiments, the online concierge system **102** also compares an identifier of a shopper who obtained the difficult to find item from the warehouse to identifiers of shoppers who were previously unable to obtain the difficult to find item from the warehouse, and prompts **820** the shopper who obtained the difficult to find item from the warehouse to provide information for finding the difficult to find item within the warehouse in response to the identifier of the shopper who obtained the difficult to find item from the warehouse differing from an identifier of a shopper who was previously unable to obtain the difficult to find item from the warehouse. This comparison of a shopper who was previously unable to obtain the difficult to find item from the warehouse to a shopper who obtained the difficult to find item from the warehouse allows the online concierge system **102** to prompt **820** different shoppers than shoppers who were unable to obtain the difficult to find item from the warehouse for information for finding the difficult to find item within the warehouse.

(64) The online concierge system **102** may maintain one or more criteria to be satisfied for a shopper who obtained the difficult to find item from the warehouse to be prompted **820** for information for finding the difficult to find item within the warehouse. In some embodiments, the criteria specify that a shopper have fully completed an order via the online concierge system **102** within a specified time interval of finding the difficult to find item within the warehouse (e.g., within 30 days of finding the difficult to find item within the warehouse), specify that a shopper have completed a minimum number of orders (e.g., 20 orders, 15 orders) via the online concierge system **102**, and that the shopper have at least a threshold amount of orders completed via the online concierge system **102** receive at least a threshold ranking from consumers for whom the orders were completed. In various embodiments, the online concierge system **102** prompts **820** shoppers who found the difficult to find item within the warehouse who satisfy at least a threshold amount of (or all of) the criteria for information for finding the difficult to find item within the warehouse, but does not prompt **820** shoppers who found the difficult to find item within the warehouse but who do not satisfy at least the threshold amount of the criteria for information for finding the difficult to find item within the warehouse. Enforcing the one or more criteria allows the online concierge system **102** to obtain information for finding the difficult to find item within the warehouse from shoppers who are likely to provide reliable or accurate information.

(65) In various embodiments, the online concierge system **102** transmits a prompt to the shopper mobile application **112** corresponding to the shopper who obtained the difficult to find item from the warehouse, and the shopper mobile application **112** displays the prompt to the shopper who obtained the difficult to find item from the warehouse. The prompt may include a message congratulating the shopper for locating the difficult to find item within the warehouse and one or more fields for providing information for finding the difficult to find item within the warehouse. In various embodiments, the information for finding the difficult to find item is a picture of a location of the difficult to find item within the warehouse, a text description of a location of the difficult to find item within the warehouse, directions for locating the difficult to find item within the warehouse, or any combination thereof.

(66) FIG. **9** shows an example interface **900** displayed by the shopper mobile application **112** displaying a prompt to a shopper to provide information for finding the difficult to find item within the warehouse. In the example shown by FIG. **9**, the interface **900** displays an image **905** of the difficult to find item. The interface **900** may also display a name of the difficult to find product or a description of the difficult to find item along with the image **905** of the difficult to find item in some embodiments. Additionally, the interface **900** displays a prompt **910** requesting the shopper provide information for finding the difficult to find item within the warehouse. The interface **900** also includes an interface element **915** that, when selected, allows the shopper to capture a photograph of a location within the warehouse where the difficult to find item was located via a client device (e.g., a mobile device) executing the shopper mobile application **112**. In the example shown by FIG. **9**, the interface **900** also includes fields **920A**, **920B** for receiving text input from the shopper. For example, the shopper enters text data describing the location in the interface where the difficult to find item was located in field **920A** and enters an aisle number or other location within the warehouse in field **920B**. Hence, interface **900** allows the shopper who found the difficult to find item within the warehouse to provide information to the online concierge system **102** allowing other shoppers to more readily locate the difficult to find item within the warehouse.

(67) Referring back to FIG. **8**, the online concierge system **102** stores **825** the information for finding the difficult to find item within the warehouse in association with an identifier of the warehouse and in association with an identifier of the difficult to find item. In some embodiments, the online concierge system also stores an identifier of the shopper who obtained the difficult to find item in association with the information for finding the difficult to find item within the warehouse. Hence, the online concierge system **102** maintains information for finding one or more difficult to find items within the warehouse, allowing the online concierge system **102** to

subsequently retrieve information for finding a difficult to find item within the warehouse.

(68) Subsequently, when the online concierge system **102** receives **830** an order to be fulfilled at the warehouse and that includes the difficult to find item within the warehouse, the online concierge system **102** retrieves **835** the information for finding the difficult to find item within the warehouse stored in association with the identifier of the warehouse and with the identifier of the difficult to find item. The online concierge system **102** displays **840** the information for finding the difficult to find item within the warehouse to a shopper fulfilling the order at the warehouse via a shopper mobile application **112** corresponding to the shopper fulfilling the order. For example, the online concierge system **102** displays an indication that information for finding the difficult to find item is available proximate to information identifying the difficult to find item in an interface of the shopper mobile application **112**. FIG. **10** shows an example interface **1000** displaying an order being fulfilled by a shopper. The interface **1000** displays different items **1005**, **1015**, **1020** included in the order, with an image and a name or a description of each item **1005**, **1015**, **1020** included in the order displayed in the interface **1000**. In the example of FIG. **10**, the difficult to find item described above in conjunction with FIG. **9** is included in the order, so image **905** of the difficult to find item is displayed in the interface **1000**. To allow the shopper fulfilling the order to more quickly locate the difficult to find item, the interface **1000** displays an indication **1010** that the online concierge system **102** maintains information for locating the difficult to find item in the warehouse proximate to the image **905** of the difficult to find item, or proximate to other information identifying the difficult to find item.

(69) Referring back to FIG. **8**, in response to receiving a selection of the indication from the shopper mobile application, the online concierge system **120** displays **840** the information for finding the difficult to find item within the warehouse, allowing the shopper fulfilling the order to leverage the stored information to more efficiently locate and retrieve the difficult to find item from the warehouse. FIG. **11** shows an example interface **1100** displayed by the shopper mobile application **112** including information for finding a difficult to find item in a warehouse. In the example shown by FIG. **11**, the interface **1100** is displayed to a shopper after the shopper selects the indication **1010** described above in conjunction with FIG. **10**. The interface **1100** includes the image **905** of the difficult to find item, and may include a name of the difficult to find item or a description of the difficult to find item in some embodiments. In the example of FIG. **11**, the interface **1100** displays an image **1105** of a location of the item in the warehouse in which the item is difficult to find and displays a text description **1110** of the location of the item in the warehouse in which the item is difficult to find. However, in other embodiments, the interface **1100** includes one of the image **1105** of the location of the item in the warehouse or the text description **1110** of the item in the warehouse; thus, the interface **1100** displays information received from a shopper who found the difficult to find item in the warehouse in based on how the shopper who found the difficult to find item provided the information to the online concierge system **102**. In some embodiments, the interface **1100** also displays a name of or other information identifying (e.g., a picture, a username, etc.) of a shopper from whom the information for finding the difficult to find item in the warehouse was received, allowing the shopper viewing the interface **1100** to determine what shopper provided the information for finding the difficult to find item in the warehouse.

(70) In the example shown by FIG. **11**, the interface **1100** also displays a positive feedback element **1115** and a negative feedback element **1120**, allowing a shopper viewing the interface **1100** to provide the online concierge system **102** with feedback regarding usefulness of the information for finding the difficult to find item in the warehouse. The shopper viewing the interface **1110** selects the positive feedback element **1115** to send an indication to the online concierge system **102** that the information for finding the difficult to find item in the warehouse was helpful in finding the difficult to find item. The shopper viewing the interface **1110** selects the negative feedback element **1120** to send a negative indication to the online concierge system **102** that information for finding the difficult to find item in the warehouse was not helpful in finding the difficult to find item. In

some embodiments, in response to receiving a threshold number of negative indications (or in response to receiving negative indications from at least a threshold number of distinct users), the online concierge system **102** stops displaying the information for finding the difficult to find item in the warehouse to users. Additionally, if a shopper sends a negative indication to the online concierge system **102** for information for finding the difficult to find item in the warehouse, the online concierge system **102** stores the negative indication in association with an identifier of the shopper and an identifier of the information for finding the difficult to find item in the warehouse (or in association with the identifier of the shopper and the information for finding the difficult to find item in the warehouse), so the online concierge system **102** does not subsequently display the information for finding the difficult to find item in the warehouse to shopper from whom the negative indication was received. In some embodiments, in response to receiving a negative indication from the shopper to whom the information for finding the difficult to find item in the warehouse was displayed, the online concierge system **102** transmits another interface or another prompt to the client device of the shopper from whom the negative indication was received that allows the user from whom the negative indication was received to modify (e.g., edit) the information for finding the difficult to find item in the warehouse. The online concierge system **102** may remove the information for finding the difficult to find item in the warehouse in response to receiving at least a threshold number of negative indications for the information for finding the difficult to find item in the warehouse or in response to receiving negative indications for the information for finding the difficult to find item from at least a threshold number of different shoppers.

ADDITIONAL CONSIDERATIONS

(71) The foregoing description of the embodiments of the invention has been presented for the purpose of illustration; it is not intended to be exhaustive or to limit the invention to the precise forms disclosed. Persons skilled in the relevant art can appreciate that many modifications and variations are possible in light of the above disclosure.

(72) Some portions of this description describe the embodiments of the invention in terms of algorithms and symbolic representations of operations on information. These algorithmic descriptions and representations are commonly used by those skilled in the data processing arts to convey the substance of their work effectively to others skilled in the art. These operations, while described functionally, computationally, or logically, are understood to be implemented by computer programs or equivalent electrical circuits, microcode, or the like. Furthermore, it has also proven convenient at times, to refer to these arrangements of operations as modules, without loss of generality. The described operations and their associated modules may be embodied in software, firmware, hardware, or any combinations thereof.

(73) Any of the steps, operations, or processes described herein may be performed or implemented with one or more hardware or software modules, alone or in combination with other devices. In one embodiment, a software module is implemented with a computer program product comprising a computer-readable medium containing computer program code, which can be executed by a computer processor for performing any or all of the steps, operations, or processes described.

(74) Embodiments of the invention may also relate to an apparatus for performing the operations herein. This apparatus may be specially constructed for the required purposes, and/or it may comprise a general-purpose computing device selectively activated or reconfigured by a computer program stored in the computer. Such a computer program may be stored in a tangible computer readable storage medium, which include any type of tangible media suitable for storing electronic instructions, and coupled to a computer system bus. Furthermore, any computing systems referred to in the specification may include a single processor or may be architectures employing multiple processor designs for increased computing capability.

(75) Embodiments of the invention may also relate to a computer data signal embodied in a carrier wave, where the computer data signal includes any embodiment of a computer program product or

other data combination described herein. The computer data signal is a product that is presented in a tangible medium or carrier wave and modulated or otherwise encoded in the carrier wave, which is tangible, and transmitted according to any suitable transmission method.

(76) Finally, the language used in the specification has been principally selected for readability and instructional purposes, and it may not have been selected to delineate or circumscribe the inventive subject matter. It is therefore intended that the scope of the invention be limited not by this detailed description, but rather by any claims that issue on an application based hereon. Accordingly, the disclosure of the embodiments of the invention is intended to be illustrative, but not limiting, of the scope of the invention, which is set forth in the following claims.

Claims

1. A computer-implemented method performed by one or more processors of an online system, the computer-implemented method comprising: training a machine-learned availability model for determining availabilities of items in warehouses, wherein training of the machine-learned availability model comprises: applying a plurality of training samples to train the machine-learned availability model, the plurality of training samples comprising previous delivery orders, wherein at least one training sample comprises a training label indicating whether an item was found at a warehouse and a plurality of characteristics associated with the item; receiving a first set of notifications regarding availabilities of a plurality of items, wherein a notification includes whether one or more items were found at the warehouse; determining, based on the first set of notifications, that a set of items were reported to be not found, the set of items being reported to be not found for at least a threshold number of times; retraining, based on the set of items, the machine-learned availability model to improve accuracy of predictions of the machine-learned availability model, wherein retraining the machine-learned availability model comprises: generating an additional set of training samples using the set of items that were not found for at least the threshold number of times, inputting the additional training samples to the machine-learned availability model for the machine-learned availability model to generate a confidence score indicative of an accuracy of a prediction of availability of one or more items in the set, responsive to the confidence score indicative of the accuracy of the prediction of availability of the specific item being below a threshold, sending instructions to indicate the availability of the one or more items need to be verified, receiving a second set of notifications indicating one or more items in the set of items were found in the warehouse, updating the additional set of training samples based on the second set of notifications, applying the additional set of training samples to the machine-learned availability model, and adjusting weights of the machine-learned availability model based on the applying of the additional set of training samples; receiving, from a user device, a selection of a particular item; applying the retrained machine-learned availability model to generate a prediction of whether the particular item is available at the warehouse; and causing to display a result based on the prediction generated by the machine-learned availability model regarding the particular item.

2. The computer-implemented method of claim 1, wherein receiving the first set of notifications regarding availabilities of the plurality of items comprises: receiving an indication that a shopper has obtained at least one of the plurality of items; determining that the obtained one of the plurality of items is a difficult-to-find item that was not found for at least the threshold number of times; updating the training samples for the machine-learned availability model to indicate that the found one of the plurality of items is a difficult-to-find item.

3. The computer-implemented method of claim 1, wherein receiving the first set of notifications regarding availabilities of the plurality of items comprises: receiving an indication that a shopper could not find one of the plurality of items; designating the one of the plurality of items as an unfound item; determining that the unfound item is a difficult-to-find item that was not found for at

least the threshold number of times; updating the additional training samples for the unfound item for the machine-learned availability model to indicate that the found one of the plurality of items is a difficult-to-find item.

4. The computer-implemented method of claim 1, wherein causing to display a result based on the prediction generated by the machine-learned availability model regarding the particular item comprises: prompting a shopper to provide information for finding the particular item in the warehouse; receiving a response from the shopper whether the shopper is able to find the particular item in the warehouse; and causing the user device to display an availability of the particular item in the warehouse.

5. The computer-implemented method of claim 4, wherein the information for finding the particular item in the warehouse comprises one or more of: a picture of a location in the warehouse of the particular item, a text description of a location in the warehouse of the particular item, or any combination thereof.

6. The computer-implemented method of claim 1, wherein applying the machine-learned availability model to generate a prediction of whether the particular item is available comprises: generating a predicted availability score associated with the particular item; comparing the predicted availability score to an availability threshold; and determining the prediction of whether the particular item is available based on the comparison.

7. The computer-implemented method of claim 1, further comprising: receiving information from a first shopper that the first shopper is able to find the particular item at the warehouse; receiving an additional order that includes the particular item; and transmitting the information to a second shopper for finding the particular item at the warehouse.

8. The computer-implemented method of claim 7, wherein receiving information from the first shopper comprises receiving a photo taken by the first shopper locating the particular item.

9. The computer-implemented method of claim 1, further comprising: receiving a threshold number of negative indications that an item is not found at the warehouse; and responsive to receiving the threshold number of negative indications, removing the item from an offering list of the online system with respect to the warehouse.

10. A non-transitory computer-readable medium configured to store code comprising instructions, wherein the instructions, when executed by one or more processors, cause the one or more processors to: train a machine-learned availability model for determining availabilities of items in warehouses, wherein training of the machine-learned availability model comprises: applying a plurality of training samples to train the machine-learned availability model, the plurality of training samples comprising previous delivery orders, wherein at least one training sample comprises a training label indicating whether an item was found at a warehouse and a plurality of characteristics associated with the item; receive a first set of notifications regarding availabilities of a plurality of items, wherein a notification includes whether one or more items were found at the warehouse; determine, based on the first set of notifications, that a set of items were reported to be not found, the set of items being reported to be not found for at least a threshold number of times; retrain, based on the set of items, the machine-learned availability model to improve accuracy of predictions of the machine-learned availability model, wherein retraining the machine-learned availability model comprises: generating an additional set of training samples using the set of items that were not found for at least the threshold number of times, inputting the additional training samples to the machine-learned availability model for the machine-learned availability model to generate a confidence score indicative of an accuracy of a prediction of availability of one or more items in the set, responsive to the confidence score indicative of the accuracy of the prediction of availability of the specific item being below a threshold, sending instructions to indicate the availability of the one or more items need to be verified, receiving a second set of notifications indicating one or more items in the set of items were found in the warehouse, updating the additional set of training samples based on the second set of notifications, applying the additional

set of training samples to the machine-learned availability model, and adjusting weights of the machine-learned availability model based on the applying of the additional set of training samples; receive, from a user device, a selection of a particular item; apply the retrained machine-learned availability model to generate a prediction of whether the particular item is available at the warehouse; and cause to display a result based on the prediction generated by the machine-learned availability model regarding the particular item.

11. The non-transitory computer-readable medium of claim 10, wherein the instruction to receive the first set of notifications regarding availabilities of the plurality of items comprises instructions to: receive an indication that a shopper has obtained at least one of the plurality of items; determine that the obtained one of the plurality of items is a difficult-to-find item that was not found for at least the threshold number of times; update the training samples for the machine-learned availability model to indicate that the found one of the plurality of items is a difficult-to-find item.

12. The non-transitory computer-readable medium of claim 10, wherein the instruction to receive the first set of notifications regarding availabilities of the plurality of items comprises instructions to: receive an indication that a shopper could not find one of the plurality of items; designate the one of the plurality of items as an unfound item; determine that the unfound item is a difficult-to-find item that was not found for at least the threshold number of times; update the additional training samples for the unfound item for the machine-learned availability model to indicate that the found one of the plurality of items is a difficult-to-find item.

13. The non-transitory computer-readable medium of claim 10, wherein the instruction to cause to display a result based on the prediction generated by the machine-learned availability model regarding the particular item comprises instructions to: prompt a shopper to provide information for finding the particular item in the warehouse; receive a response from the shopper whether the shopper is able to find the particular item in the warehouse; and cause the user device to display an availability of the particular item in the warehouse.

14. The non-transitory computer-readable medium of claim 13, wherein the information for finding the particular item in the warehouse comprises one or more of: a picture of a location in the warehouse of the particular item, a text description of a location in the warehouse of the particular item, or any combination thereof.

15. The non-transitory computer-readable medium of claim 10, wherein the instruction to applying the machine-learned availability model to generate a prediction of whether the particular item is available comprises instructions to: generate a predicted availability score associated with the particular item; compare the predicted availability score to an availability threshold; and determine the prediction of whether the particular item is available based on the comparison.

16. The non-transitory computer-readable medium of claim 10, wherein the instructions, when executed, further cause the one or more processors to: receive information from a first shopper that the first shopper is able to find the particular item at the warehouse; receive an additional order that includes the particular item; and transmit the information to a second shopper for finding the particular item at the warehouse.

17. The non-transitory computer-readable medium of claim 16, wherein the instruction to receive information from the first shopper comprises instructions to receive a photo taken by the first shopper locating the particular item.

18. The non-transitory computer-readable medium of claim 10, wherein the instructions, when executed, further cause the one or more processors to: receive a threshold number of negative indications that an item is not found at the warehouse; and responsive to receiving the threshold number of negative indications, remove the item from an offering list of an online system with respect to the warehouse.

19. A system comprising: one or more processors; and memory configured to store code comprising instructions, wherein the instructions, when executed by one or more processors, cause the one or more processors to: train a machine-learned availability model for determining

availabilities of items in warehouses, wherein training of the machine-learned availability model comprises: applying a plurality of training samples to train the machine-learned availability model, the plurality of training samples comprising previous delivery orders, wherein at least one training sample comprises a training label indicating whether an item was found at a warehouse and a plurality of characteristics associated with the item; receive a first set of notifications regarding availabilities of a plurality of items, wherein a notification includes whether one or more items were found at the warehouse; determine, based on the first set of notifications, that a set of items were reported to be not found, the set of items being reported to be not found for at least a threshold number of times; retrain, based on the set of items, the machine-learned availability model to improve accuracy of predictions of the machine-learned availability model, wherein retraining the machine-learned availability model comprises: generating an additional set of training samples using the set of items that were not found for at least the threshold number of times, inputting the additional training samples to the machine-learned availability model for the machine-learned availability model to generate a confidence score indicative of an accuracy of a prediction of availability of one or more items in the set, responsive to the confidence score indicative of the accuracy of the prediction of availability of the specific item being below a threshold, sending instructions to indicate the availability of the one or more items need to be verified, receiving a second set of notifications indicating one or more items in the set of items were found in the warehouse, updating the additional set of training samples based on the second set of notifications, applying the additional set of training samples to the machine-learned availability model, and adjusting weights of the machine-learned availability model based on the applying of the additional set of training samples; receive, from a user device, a selection of a particular item; apply the retrained machine-learned availability model to generate a prediction of whether the particular item is available at the warehouse; and cause to display a result based on the prediction generated by the machine-learned availability model regarding the particular item.

20. The system of claim 19, wherein the instruction to receive the first set of notifications regarding availabilities of the plurality of items comprises instructions to: receive an indication that a shopper has obtained at least one of the plurality of items; determine that the obtained one of the plurality of items is a difficult-to-find item that was not found for at least the threshold number of times; update the training samples for the machine-learned availability model to indicate that the found one of the plurality of items is a difficult-to-find item.
